Northeastern University, Khoury College of Computer Science

CS 6120 Natural Language Processing — Lab T14

April 13, 2026 (Week 12)

Agents with an Execution Environment

Objective

The goal of this lab is to set up a self-hosted agent using **OpenClaw**. You will provision a virtual environment, install the OpenClaw execution framework, and bridge it to a Telegram account to allow for remote agentic interactions.

Step 1: Telegram Account Setup

To interact with your agent via messaging, you need an active Telegram account.

- Install the Telegram app on your mobile device or desktop.
- Register using a valid phone number.

Step 2: Provisioning the Execution Environment

OpenClaw requires a Linux environment to run commands and manage tools.

1. Navigate to the **Google Cloud Console (GCP)**.
2. Create a new Virtual Machine (VM) instance (e2-medium is recommended).
3. Ensure you have SSH access to the instance and that outgoing traffic is allowed.

Step 3: Installing OpenClaw

Run the following command in your VM terminal to install the OpenClaw CLI:

```
curl -fsSL https://openclaw.ai/install.sh | bash
```

Step 4: API Configuration & Custom JSON Setup

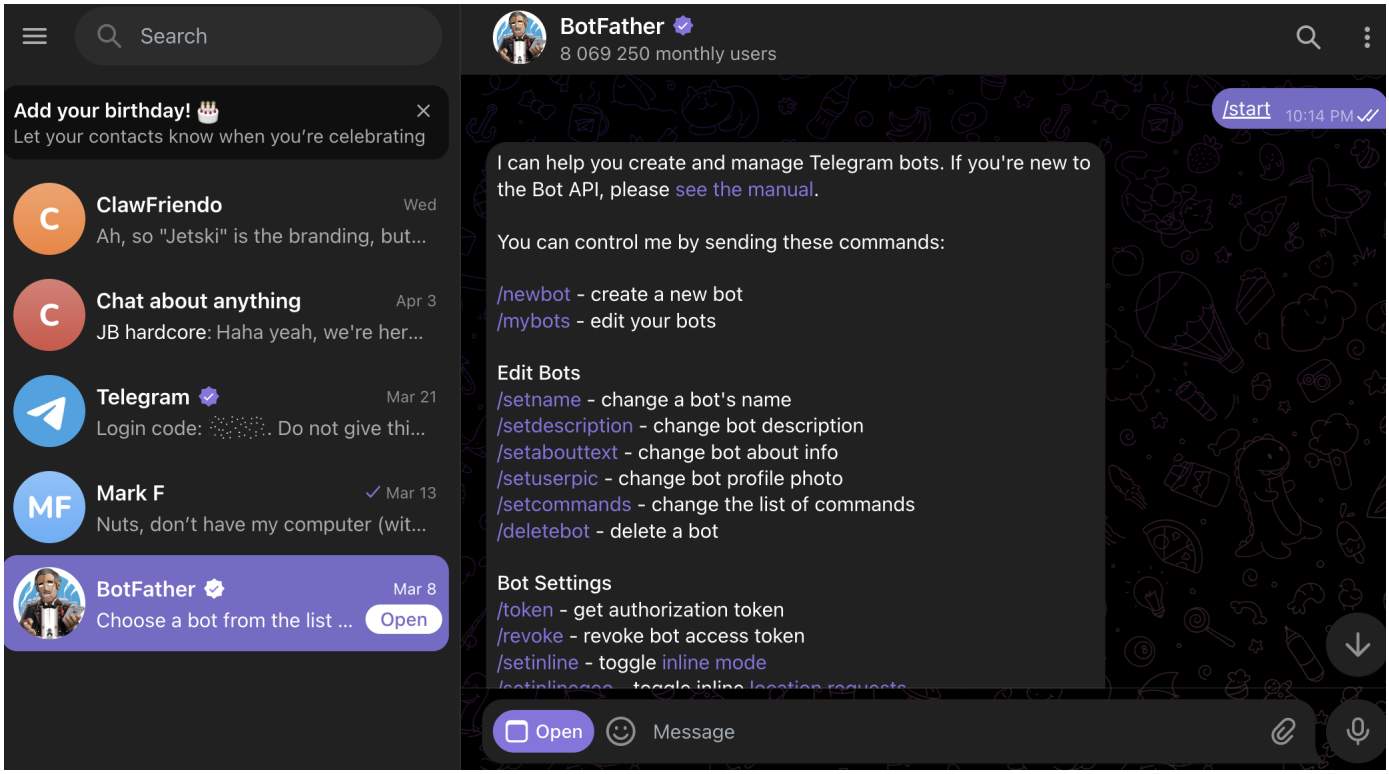
OpenClaw defaults to OpenAI. However because we have Google credits and for better stability and feature support, we will configure it to use Gemini via a custom JSON configuration.

1. **Obtain API Keys:** The primary keys that you can use are the free ones from AI Studio in Google.
   a) Log in to the **Google AI Studio** console with your project and create a new Secret Key: Google's AI Studio
   b) You may also need additional keys. There are a handful of sites that provide free API keys for limited usage. One that you can try is: OpenRouter.AI
2. **Review Reference Configuration:** The main file that you create when installing OpenClaw is the json configuration file. You'll need to edit it for any new tool that you'd like to provide it.
   - The file is located at: `.openclaw/openclaw.json`
   - Feel free to take a look at my openclaw.json file. (Don't copy it; it's only a reference.)
3. **Edit the Credentials:** Open the file to insert your specific API keys:
   `nano .openclaw/openclaw.json`
4. **Update the Model Provider:** Inside the file, ensure the `"provider"` is set to `"google"` and paste your key into the `"api_key"` field.
5. **Verify Schema:** OpenClaw is strict about JSON formatting. You can verify your file is valid by running: `openclaw config doctor`

Step 5: Connecting Messaging Channels via @BotFather

To link OpenClaw to Telegram, you must first create a bot identity:

1. Open Telegram and search for **@BotFather**.
2. Send the command `/newbot` and follow the prompts to name your bot.
3. Copy the **API Token** provided by @BotFather.
4. In your VM terminal, run: `openclaw channels add`
5. Select **Telegram** and paste your API Token when prompted.
6. Verify the connection with: `openclaw channels list`



Exercise

Once connected, message your new bot on Telegram and ask it: *"Check the disk usage on my server and tell me if I'm running out of space."* Then, join ask course staff to join the group, noting the name of your ClawBot.